KENDRIYA VIDYALAYA SANGATHAN
[GUWAHATI REGION]
SESSION ENDING EXAMINATION 2016
SUBJECT : BIOLOGY
CLASS–XI
(SOLVED PAPER)

Time : 3 Hours 30 Min. OTBA **Max. Marks : 70**

General Instructions :

1. *The question paper comprises of five sections A, B, C, D, E & F.*
2. *All questions are compulsory.*
3. *There is no overall choice however; internal choice has been provided in one question of 2 marks, one question of 3 marks and all the questions of five marks category. Only one option in such question is to be attempted.*
4. *Questions 1 to 4 in Section A are very short questions of one mark each. These are to be answered in one sentence each.*
5. *Questions 5 to 9 in Section B are short questions of two marks each. These are to be answered in approximately 20-30 words each.*
6. *Questions 10 to 18 in Section C are questions of three marks each. These are to be answered in approximately 30-50 words each.*
7. *Question 19 in Section D is Value based question and is of 4 marks.*
8. *Questions 20 to 22 in Section E are questions of five marks each. These are to be answered in approximately 80-120 words each.*
9. *Questions 23 to 24 in section F is based on OTBA of 10 marks.*

SECTION - A

1. What do you mean by epipetalous ? 1

2. Name two types of parenchyma present in leaf. 1

3. What is Glycosidic Bond ? 1

4. Mention two significance of Mitosis. 1

SECTION - B

5. Differentiate G_1 and G_2 phase of cell cycle. 2

6. Define the term isotonic solution. 2

7. What are uricotelic animals ? Give two examples. 2

8. Write the floral formula for the family Fabaceae. 2

9. Differentiate between apoplast and symplast pathway. 2

OR

What is water potential ? What is the relationship between water potential with Solute potential and Pressure potential ?

SECTION - C

10. Draw the diagram of V.S of maize seed and label six parts. 3

11. RNA and DNA are the basic genetic materials of living beings. Write three points of difference between them. 3

12. In certain plants growing in temperate region suffer from photorespiratory loss. How do they overcome it ? Mention the anatomical adaptation. Mention the first stable CO_2 fixation product and the enzyme responsible for it. 3

13. What is non-symbiotic N_2 fixation ? Give examples of some microorganism. 3

14. What is non-cyclic photophosphorylation ? Describe the process. Why is the process referred to as non-cyclic ? 3

OR

What is cyclic photophosphorylation ? Describe the process using a flow diagram. Why is the process referred to as cyclic ?

15. Draw a sketch of human ear and label various parts. 3

OR

Draw a labeled diagram of human eye and label various parts.

16. How many ATP are produced during cellular respiration during – Glycolysis, Oxidative decarboxylation and Kreb's Cycle ? What is the net production of ATP ? Define Respiratory Quotient. 3

17. What is placentation ? Mention the types. Describe any one with typical diagram. 3

18. Differentiate between
(a) Early wood and late wood
(b) Heart wood and sap wood 3

SECTION - D

19. In a class seminar Ashok described about the role, distribution and characteristics of vascular bundle in monocot and dicot plants. Arun was under impression that the vascular bundle distribution is same in both monocot and dicot. To explain, it Ashok showed the T.S. of monocot and dicot stem to Arun under microscope.
(a) What character did Ashok demonstrate ?
(b) What is vascular bundle according to you ?
(c) What is the difference between the distribution of vascular bundle in monocot and dicot stem ? 4

SECTION - E

20. Compare the characters of Chordates and Non-chordates. 5

OR

Classify sub-phylum Vertebrata using flow diagram.

21. Draw a labelled diagram of Alimentary canal of cockroach.
What is the function of crop and gizzard in the digestive system of cockroach ? 5

OR

Draw a labelled diagram of male reproductive system of cockroach.
What is the function of testes in the reproductive system of cockroach ?

22. Explain the process of urine formation in Human being. Explain briefly the various steps . 5

OR

(A) Mention the function of the following in regulation of kidney function
(i) ADH, (ii) ANF, (iii) Renin
(B) How is the heart activity regulated in human body ?

SECTION - F (OTBA)

Read the section after the Text material provided under OTBA

23. What is air pollution ? Mention some primary and secondary air pollutants. Mention the effect of air pollution on health. 5

24. What is Pneumoconiosis ? Mention its types. Write briefly how it can be prevented and controlled ? 5

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SOLUTIONS

SECTION - A

1. **Epipetalous –** When stamens are fused with petals, the condition is called epipetalous. Eg. Solanum, Petunia. 1

2. **(i)** Palisade parenchyma
(ii) Spongy parenchyma ½ + ½

3. **Glycosidic Bond –** It is a bond formed during the condensation of monosaccharides for the formation of oligosaccharides and polysaccharides. Eg. Cellulose. 1

4. **Two significances of Mitosis :**
(i) Mitosis is essential for growth and development of multicellular organisms.
(ii) Mitosis is the method of multiplication of unicellular organisms. ½ + ½

SECTION - B

5. **Differentiation between G_1 and G_2 phase is as follows :**

S. No.	G_1	G_2
(i)	This stage starts from the birth of the daughter cell and ends upto the time when the cell begins to synthesise DNA.	This stage starts from the end of S-phase upto the beginning of M-phase.
(ii)	RNA and proteins are synthesized.	In G_2 phase synthesis of DNA stops. Formation of RNA and protein continue.

1 + 1

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SESSION ENDING EXAMINATION 2016
SUBJECT : BIOLOGY
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Time : 3 Hours **Max. Marks : 70**

General Instructions :

1. *There are a total of 26 questions and five sections in the question paper. All questions are compulsory.*
2. *Section A contains question numbers 1 – 5, very short answer type question of 1 mark each.*
3. *Section B contains question numbers 6 – 9, short answer type I questions of 2 marks each.*
4. *Section C contains question numbers 10-20, short answer type II question of 3 marks each.*
5. *Question numbers 21 is a value based questions, of 4 marks.*
6. *Section D contains question numbers question 22 & 23, of 5 mark each.*
7. *Question 24, 25 & 26 in section E is based on OTBA of 10 marks.*
8. *There is no overall choice. However, an internal choice has been provided in one question of two marks, in one question of 3 marks and in all the three questions of 5 marks. An examinee is to attempt only one of the choices in such questions.*

SECTION - A

1. What is metamerism ? 1
2. What type of Phyllotaxy is seen in Alstonia and Guava ? 1
3. What is G_0-phase ? 1
4. How many molecules of ATP and how many molecules of $NADPH+H^+$ are spent to fix three molecules of CO_2 ? 1
5. What happen when the secretion of hormone ADH is inhibited ? 1

SECTION - B

6. Why is ATP called as the coin of the energy ? 2
7. Write two differences between anaerobic respiration and fermentation. 2
8. Name two leucocytes which are phagocytic. 2
9. Bile does not contain any digestive enzymes, yet it is essential for digestion. Why ? 2

OR

Salivary amylase and pancreatic amylase both work on starch but still both are different to each other. Write two differences between them ?

SECTION - C

10. Differentiate between green algae, brown algae and red algae in respect of pigments and reserve food material. 3
11. (a) How important is the presence of air bladder in Class Osteichthyes ? 3
    (b) Expand ICZN.
12. Describe the corolla of family Fabaceae ? 3
13. (a) Name the connective tissue that lacks fibre in its matrix.
    (b) Write two differences between male and female cockroach. 3
14. (a) What is periderm ? How does periderm formation take place in dicot ?
    (b) How are vascular bundles arranged in monocot root ? 3
15. Describe the primary, secondary and tertiary structure of protein ? 3
16. Name and write the function of different kinds of leucoplast in plant cell. 3
17. State the effect of chemicals on enzyme action with suitable example. 3
18. What is oxidative decarboxylation ? What is its significance for Kreb's cycle ? 3
19. Write one function of sulphur, potassium and iron ? 3
20. What are the components of limbic system ? What is its function ? 3

OR

How nerve impulses are conducted in non-myelinated nerves ?

21. Muscles are the specialized tissue of mesodermal origin and show specific features responsible for movement. The muscles are of three types, based on their location, appearance and nature of regulation of their activities.

(a) Name the three types of muscles based on their location.

(b) Why are visceral muscles called as smooth muscles as well as involuntary muscles ?

(c) What value is imparted to you by the muscle functioning ? 4

SECTION - D

22. What is cell cycle ? Describe the phases of cell cycle with suitable diagram. 5

OR

Describe prophase-I in detail with diagrams.

23. (a) Explain the process of nitrogen fixation in leguminous plant. Where does this process occur in plant ? 5

(b) All minerals cannot be absorbed by passive method of absorption. Why ?

OR

Explain the photochemical phase of photosynthesis. What is the light reaction product ?

SECTION - E (OTBA)

24. In which respect human brain differ from small intestine which is considered as second brain. Write two point differences ? 3

25. What is Probiotics ? What is its role in maintaining health ? 2

26. Name two chemicals which produce PAN. Write the full form of PAN. Write three methods to reduce air pollution. 5

■■

SOLUTIONS

SECTION - A

1. In some animals, the body is externally and internally divided into segments with a serial repetition of at least some organs. eg. Earthworm, Neries etc. 1

2. (i) **Alstonia :** Whorled phyllotaxy

   (ii) **Guava :** Opposite and decussate ½ + ½

3. G_0-phase is the arrest of cell cycle and onset of differentiation. 1

4. Fixation of three molecules of CO_2 uses 9 ATP and 6 NADPH molecules in C_3 plants.

   In C_4 plants for fixation of 3 CO_2 molecules, 15 ATP and 6 NADPH energy is required. ½ + ½

5. Urine output is inversely proportional to ADH. If ADH secretion will reduce it increases the urine output and the osmolarity of blood will also increases. 1

SECTION - B

6. **ATP (Adenosine triphosphate) :** It is an instant store house as well as source of energy. ATP is found in all living cells. It is also mobile within the cell. Therefore, ATP can pick up energy where it is being liberated and supply the same where it is consumed. So ATP is called universal energy carrier or coin of the energy. 2

7. **Differentiation between anaerobic respiration and fermentation :**

S. No.	Anaerobic Respiration	Fermentation
(i)	It is the type of respiration.	It is energy controlled breakdown and transformation of organic nutrients.
(ii)	It is an intracellular process.	It is both an intracellular or extracellular process.

1 + 1

8. Monocytes and Neutrophils. 1 + 1

9. Importance of bile in digestion :
   (i) Bile neutralizes acidic medium.
   (ii) Bile is germicidal.
   (iii) Bile emulsifies fat.
   (iv) Bile activates lipase. 2

OR

Differentiation between salivary amylase and pancreatic amylase :

S. No.	Salivary amylase	Pancreatic amylase
(i)	Salivary amylase is produced in the salivary glands.	Pancreatic amylase is produced in the Pancreas.
(ii)	Salivary amylase starts digestion of simple carbohydrates in the mouth.	Pancreatic amylase acts more on complex carbohydrates in the small intestine.

1 + 1

SECTION - C

10. Difference between red, brown and green algae in respect of pigments and reserve food material :

S. No.	Green algae	Brown algae	Red algae
(i)	Pigments – chlorophyll a, chlorohyll b, carotenes and xantholphylls are present.	In brown algae fucoxanthin, Chlorophyll a and c type are present.	Red in colour due to presence of phycoerythrin. As well as Chlorophyll a and d type are present.
(ii)	The reserve food is starch.	The reserve food is laminarian starch.	Food is stored in the form of floridean starch.

3

11. (a) Class osteichthyes (Bony fishes) have a sac-like outgrowth, the swim bladder is also called air bladder, arising from the dorsal wall of the oesophagus, which is air filled organ, used to maintain balance and to swim up

and down. It also enables the fish to stay at particular depth without spending energy. In some fishes it helps in respiration such as Heteropneustes.

(b) International Code of Zoological Nomenclature. (ICZN) **2 + 1**

12. **Corolla of family Fabaceae :** Corolla 5, polypetalous, aestivation descending or vexillary imbricate, pilionaceous with five unequal petals-posterior largest petal called standard which overlaps two smaller lateral petals called wings, the latter overlap a boat-shaped structure called keel, which is formed by the two anterior petals fused lightly on the anterior side, lateral and anterior petals are scale-like in Erythrina indica. **3**

13. **(a)** Vascular tissue – Blood and lymph.

(b) **Difference between male cockroach and female cockroach :**

S. No.	Male Cockroach	Female Cockroach
(i)	It has anal styles at posterior end of abdomen.	Anal style are absent.
(ii)	Male genital aperture lies below anus.	Female genital aperture lies on 8^{th} sternum.

1 + 2

14. **(a)** **Periderm :** The cork, cork cambium and secondary cortex collectively form a protective tissue called periderm.

In order to provide for increase in girth and prevent harm on the rupturing of the outer ground tissues due to the formation of secondary vascular tissues, dicot stems produce a cork cambium or phellgen in the outer cortical cells. Phellogen cells divide on both the outer side as well as the inner side to form secondary tissue. The secondary tissue produced on the inner side of the phellogen is parenchymatous . It is called secondary cortex or phelloderm. Phellogen produces cork or phellem on the outer side. The phelloderm, phellogen and phellem together constitute the periderm.

(b) The vascular bundles are arranged in the form of ring around the central pith. The xylem bundles are exarch, *i.e.*, protoxylem lies towards the outer side while the metaxylem faces inwards. Phloem bundles alternate with xylem bundles. **2 + 1**

15. **Primary Structure :** The linear sequence of amino acids units in a polypeptide chain is called the primary structure of protein molecule.

Ribonuclease an enzyme, is a protein with primary structure only.

Secondary Structure : The polypeptides chain form helical coils or plated sheets to form a 3-dimensional structure. This is called as secondary structure.

It may have 3 types of helices.

(i) **α-helix :** The coils of helix are held together by hydrogen bonds between amino acids.

(ii) **β-helix :** More than one chain lie side by side joined together by hydrogen bonds between the carboxyl group of amino acids.

(iii) **Collagen proteins :** It is a right handed super helix formed by the twisting of three helices of polypeptide chains intertwined with each other.

Tertiary Structure : The tertiary structure of protein brings distant amino acid side chains nearer to form active sites of enzyme proteins. The tertiary structure is maintained by weak bonds, such as hydrogen, ionic and hydrophilic-hydrophobic bonds, formed between one part of a polypeptide and another. The biological activity of a protein molecule depends largely on the specific tertiary structure. Tertiary structure is found in globular proteins. **3**

16. **Leucoplasts :** They are the colourless plastids and do not contain any pigment. They usually perform the function of storage of reserve food material. They are classified into following three types :

(i) **Amyloplasts :** Amyloplasts are generally found in tubers, endosperms and cotyledons. They store starch as reserve food material.

(ii) **Aleuronplasts :** Aleuroplasts are found in the seeds of Ricinus, maize etc. They store proteins as a reserve food material.

(iii) **Elaioplasts :** Elaioplasts are found in most monocotyledonous plants. They store oil as reserve food material. **$1 \times 3 = 3$**

17. The enzymes are proteins having high molecular weight. Enzymes are highly specific in their action. Enzymes action can be inhibited by certain chemicals. It is called enzyme inhibition, *e.g.* Non-competitive inhibition.

Non-competitive inhibition : Some chemicals act like poison and destroy the enzyme structure. Non-competitive inhibitors do not resemble the substrate in structure. They inhibit the enzyme action by binding to the enzyme at the place other than the active site, *e.g.* cyanide inhibits the mitochondrial enzyme cytochrome oxidase which is necessary for respiration. **3**

18. **Oxidative decarboxylation :** Pyruvic acid in molecules enter in the mitochondria where each of them is converted into two carbon atoms-acetic acid. One carbon is released as CO_2. The removal of carbon dioxide from pyruvic acid is called decarboxylation.

Oxidative decarboxylation plays very significant role for Kreb's cycle. This step is called link reaction or transition reaction or gateway step as it links glycolysis with krebs cycle. 3

19. Functions of sulphur, potassium and iron :

Sulphur : Chlorophyll formation and nodule formation in legumes.

Potassium : It plays important role in energy transfer and cell division.

Iron : It involves in development of chloroplast, chlorophyll and other pigments. $1 \times 3 = 3$

20. **Components of limbic system :** Certain components of the cerebrum and diencephalon constitute the limbic system. Its main components are the following :

(a) **Hippocampus :** Its shape roughly resemble the sea horse. It is located inside the temporal lobe.

(b) **Amygdala :** It is almond shaped and is located in the tip of temporal lobe.

(c) **Septal nuclei :** These are located within the septal area formed by the regions under corpus callosum and the paraterminal gyrus.

(d) **Mammillary bodies :** These are present behind the infundibulum. 3

Function of limbic system :

(a) It controls emotional behaviour expressed in the form of joy, sorrow, fear, fight etc.

(b) It controls Rood habits necessary for survival of the individual.

(c) It also controls sex behaviour necessary for survival of the splecies.

OR

Transmission of nerve impulse through non-myelinated nerves is described as :

(i) In resting stage, the Na^+ ions are pumped out from the axoplasm. This process is called sodium pump. It needs energy to work normally.

(ii) Thus axon membrane is electronegative inside and electropositive outside, this is called resting potential. In this state the nerve is said to be in polarized state.

(iii) When the nerve fibre is stimulated, it cause (electrochemical disturbance in the nerve fibre) a change in potential. This change is called action potential.

(iv) Thus, Na^+ migrate into the axoplasm and Cl^- ions diffuse out. Thus negativity is increased outside and positivity is increased inside. This is called depolarisation.

(v) Depolarisation progresses along the nerve fibre in both directions from the point of stimulus.

About 5 cms behind the moving wave of depolarisation, repolarisation begins. It is just a reverse of depolarization (Na^+ ions pumped out). 3

21. **(a)** Three types of muscles :

(i) Skeletal muscles, (ii) Smooth muscles, (iii) Cardiac muscles.

(b) Visceral muscles are called smooth muscles or involuntary muscles because action of these muscles is controlled by autonomic nervous system and hence they are not under the control of individual's will.

(c) Muscles plays very important role in body functioning. Muscles are responsible for the locomotion and movements of various body parts. $1 \times 3 = 3$

SECTION - D

22. The cycle of cell, growth and division is called cell cycle. The cell cycle is divided into four phases :

(i) G_1-phase or presynthetic phase.

(ii) S-phase or synthetic phase

(iii) G_2-phase or pre-mitotic gap phase

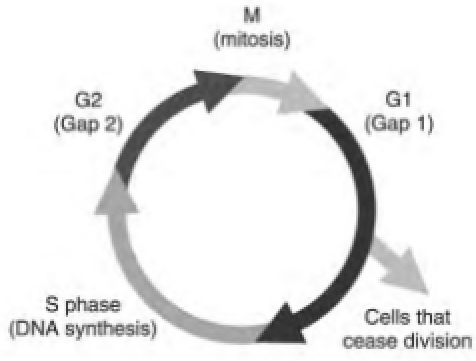
(iv) Cell division phase or mitotic phase

(i) **G_1-phase :** This stage starts from the birth of the daughter cell and ends upto the time when the cells begin to synthesise DNA. It is the phase of initial growth in which RNA and proteins are synthesized. Its duration varies from 6 to 8 hours.

(ii) **S-phase :** It is the period which starts from the time when the cell begins to synthesise DNA upto a stage when the DNA is replicated and the chromosomes are completely duplicated. It lasts for 5-6 hours.

(iii) **G_2-phase :** During this phase the centrioles, mitochondria, golgi bodies and other cytoplasmic organelles are doubled. Proteins are synthesized for the formation of astral rays and spindle and energy is stored for the cell division or M-phase. The duration of this stage starts from the end of S-phase upto the beginning of M-phase. It lasts for 4-8 hours.

(iv) **Mitotic phase :** It is the phase when the cell enters the prophase stage of cell division and at the end, gives rise to two daughter cells. It lasts for one hour.



Cell Cycle 5

OR

Prophase-I : It is the longest phase and consists of five sub-stages :

(i) Leptotene

(ii) Zygotene

(iii) Pachytene

(iv) Diplotene

(v) Diakinasis

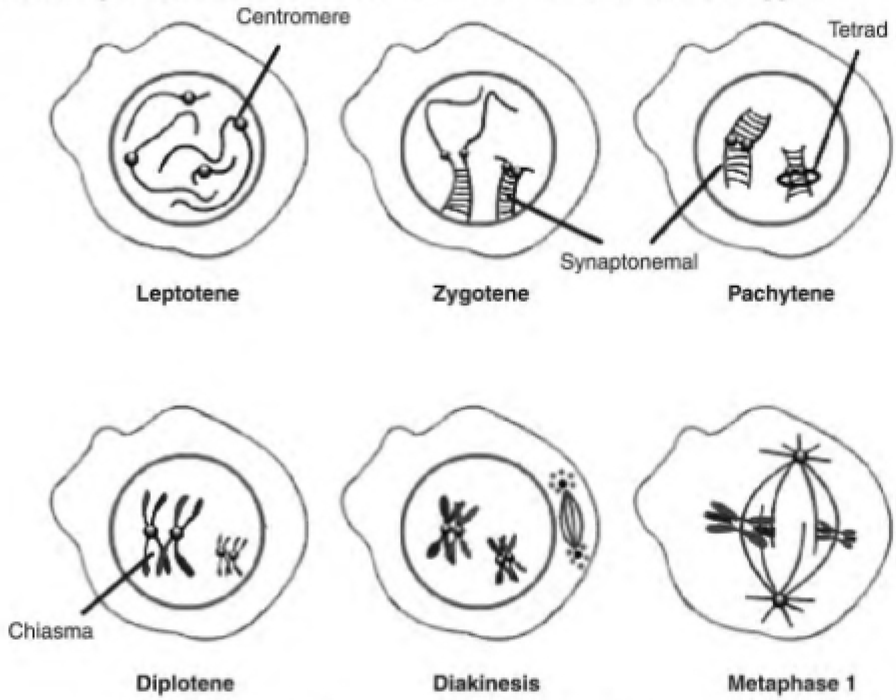
(i) Leptotene : Volume of nucleus increases and homologous chromosome become distinct as coiled thread like structure.

(ii) Zygotene : Homologous chromosomes undergo pairing (Synapsis). Each pair is called a bivalent.

(iii) Pachytene : Each chromosome of the pair splits into two chromatids thus forming a tetravalent stage. Exchange of genetic material (crossing over) takes place between inner two of the four chromatids.

(iv) Diplotene : X-shaped structures called chiasmata appears at the points where crossing over has occurred. The process of terminalization starts by which homologous chromosomes start separating.

(v) Diakinesis : Chiasmata move from centromere to the end of chromosome (terminalization) and the chromosome get thicker and shortened. Nucleolus and nuclear membrane disappears.



5

Diagram showing sub-stages of prophase-I

23. **(a)** Rhizobium is nitrogen fixing bacterial symbiont of legumes. Rhizobium live free as aerobes in the soil but are unable to fix nitrogen. They develop the ability to fix nitrogen only as a symbiont when they become anaerobic. Rhizobium is rod-shaped bacterium and it is most important for crop lands because it is associated with pulses and other legumes of family fabaceae *e.g.* chick pea or Gram. They are unable to fix nitrogen by themselves. Roots of a legume secrete chemical attractants (flavonoids and betaines). Bacteria collect over the root hairs, release nod factors that cause curling of root hairs around the bacteria, degradation of cell wall and formation of an infection thread enclosing the bacteria. Infection thread grows along with multiplication of bacteria. The infected cortical cells dedifferentiate and start dividing. It produces swelling or nodules.

Mechanism of nitrogen fixation : Nitrogen fixation require :

(i) a reducing power like NADPH, $FMNH_2$

(ii) a source of energy like ATP

(iii) enzyme dinitrogenase

(iv) compound for traping ammonia formed by the reduction of dinitrogen

Enzyme nitrogenase has iron and molybdenum. Both of them take part in attachment of a molecule of nitrogen. Bond between the two atoms of nitrogen become weakened by their attachment to the metallic components. The weakened molecule of nitrogen is acted upon by hydrogen from a reduced coenzyme. It produces dimide (N_2H_2), hydrazine (N_2H_4) and then ammonia ($2NH_3$). Ammonia is not liberated. The reaction between ammonia and organic acids give rise to amino acids.

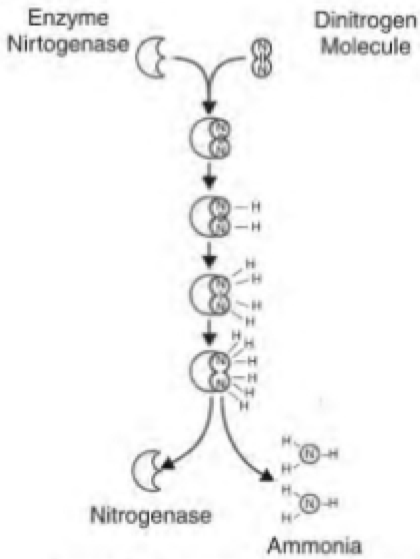


Diagram showing nitrogen fixation

All these reactions takes place in the root nodules of legume plants. 3

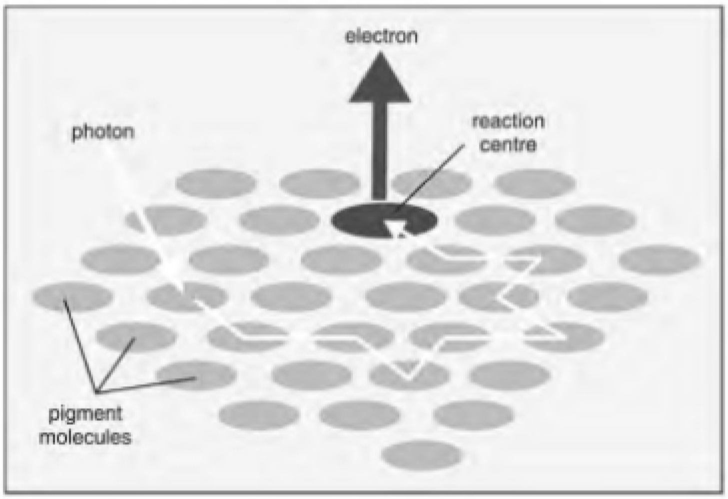
(b) All minerals cannot be absorbed by passive method of absorption because the polar molecules cannot pass through the nonpolar lipid bilayer, they require a carrier protein of the membrane to facilitate their transport across the memberane.

A few ions or molecules are transport across the membrane against their concentration gradient *i.e.* from lower to the higher concentration such transport is an energy dependent process and is called active transport. 2

OR

The photochemical phase : This phase of photosynthesis occurs in thylakoids of grana. This involve all those events which require sunlight.

A photon absorbed by the molecules of the trapping or harvesting zone of PS I and PS II pass their energy to their respective reaction centre.



For pigment system I the light energy is gathered by chlorophyll a, and possibly by chlorophyll b, while for PS II light energy is gathered by chlorophyll a, chlorophyll b and phycobillins.

Arnon et. al (1954) discovered that in addition to carrying out Hill reaction the chloroplast also synthesize ATP in presence of light from ADP and inorganic phosphate. This process is called as photophosphorylation. **5**

SECTION - E

24. Small intestine is considered as second brain because a big part of our emotions are probably influenced by the nerves in our gut. Small intestine or Second brain differs from the brain in the following way :

S. No.	Brain	Small Intestine
(i)	Brain is directly involved in thought process of the Body.	The second brain does not help with the great thought process.
(ii)	Brain regulates the functioning of various glands.	It does not involve in regulation of glands.

3

25. Probiotics are live bacteria and yeast that are good for your health, specially your digestive system. We usually think of bacteria as something that causes disease. But your body is full of bacteria, both good and bad. Probiotics are often called good or helpful bacteria because they help keep your gut healthy.

Role of probiotics in health :

(a) When you lose good bacteria in your body (like after you take antibiotics) probiotics can help replace them.

(b) They can help balance your good and bad bacteria to keep your body working like it should. **2**

26. Two chemicals producing PAN are :

(i) Aldehyde, Ketones

(ii) Dicarbonyl Compounds

PAN : Peroxy-acyl Nitrates

Three methods to reduce air pollution :

(i) Zoning of industries away from human settlement for dispersing pollution source.

(ii) Destroying pollutants by thermal or catalytic combustion.

(iii) Changing pollutants to less toxic forms. **5**

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SESSION ENDING EXAMINATION 2017
SUBJECT : BIOLOGY
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Time : 3 Hours **Max. Marks : 90**

General Instructions :

*(i) Answers to questions carrying **1** mark may be answered in one word to one sentence.*

*(ii) Answers to questions carrying **3** marks may be answered in about **50-75** words.*

*(iii) Answer to questions carrying **4** marks may be answered in about **150** words.*

*(iv) Answers to questions carrying **6** marks may be answered in about **200** words.*

(v) Attempt all parts of a question together.

1. What is the nature of the cell wall? **1**

2. Name the substance that fixes the brown algae to the substrate. **1**

3. Name the site of synthesis of glycoproteins and glycolipids in a cell. **1**

4. How many types of amino acids found to occur in proteins? **1**

5. What is the value of water potential of pure water? **1**

6. Bring out the differences between chlorophyceae and phaeophyceae. **2**

7. Give the location of hepatic caecae in a cockroach. What is their function? **2**

8. What are kinetochores? What is their function? **2**

9. *'Omnis cellula-e-cellula'*. Who gave this statement and what does it mean? **2**

OR

Describe the role of haemoglobin in the transport of respiratory gases. **2**

10. Write differences between arithmetic and geometric growth rate. **2**

11. Represent diagrammatically the apoplastic and symplastic pathways of water transport in plants. **3**

12. What is meant by flux? Describe its two kinds.

13. Represent systematically the process of ATP synthesis through chemiosmosis in chloroplast. **3**

14. Give the structural formula of (a) adenine (b) adenosine (c) adenylic acid

15. Draw a well labeled diagram of mitochondria. **3**

16. Where does mitosis take place in plants and animals? What is its significance in multicellular organisms? **3**

17. Represent diagrammatically a pinnately compound leaf and a palmately compound leaf. Name one example of each. **3**

18. What is periderm? How does periderm formation takes place in dicot stem? **3**

19. Draw a well labeled diagram of alimentary canal of a cockroach. **3**

20. Who proposed the five kingdom classification? Name the five kingdoms. **3**

OR

How are the animals of Arthropoda different from those of Mollusca? **3**

21. What is exophthalmic goitre? Mention any four symptoms of this disorder. **3**

22. Write short notes on hind brain. **3**

23. Leena and Rita are final year BSc. students. Suddenly their classmate Arun's mother needed blood transfusion. She is B positive. Leena and Rita made all efforts to find suitable blood donor from their college to help Arun.

(a) What blood group/groups can be transfused by Arun's mother.

(b) Name the blood group known as universal donor.

(c) Name the blood group referred to as universal acceptor.

(d) What values are shown by Leena and Rita? **4**

24. Where does non cyclic photophosphorylation take place? Describe this process with the help of its schematic representation. Why is this process called so? **5**

OR

Where does glycolysis occur in cell? Describe the sequence of reactions in it. Mention the end product. **5**

25. Suggest five measures that can be proposed at global level for the prevention and control of diabetes and its complications. **5**

26. Mention the schemes implemented by government of India to tackle malnutrition and health problems in children. **5**

SOLUTIONS

1. The plant cell wall is a complex matrix of linked polysaccharides such as cellulose and pectin, forming a thick semi-permeable rigid barrier outside the plasma membrane.

2. Algin **1**

3. Golgi Apparatus **1**

4. 20 **1**

5. Zero **1**

6.

S. No.	Phaeophyceae	Chlorophyceae
1.	Marine forms.	Chiefly fresh water in nature.
2.	Unicellular forms do not exist.	Unicellular species are more.
3.	Thylakoids occur in groups of three.	Stacked in groups of 2-20.
4.	Chlorophyll 'a' and 'c' present.	Chlorophyll 'a' and 'b' is present.
5.	Fucoxanthin is present.	Fucoxanthin is absent.
6.	Reserve food is laminarin.	Reserve food is starch.

(Any two) 1 × 2 = 2

7. Hepatic or gastric caecae are 6-8 narrow and hollow blind tubules called is present at the junction of foregut and mid gut. The hepatic caecae are similar to vertebrate liver, secretes digestive juices and help in the digestion. **1+1=2**

8. The kinetochore is a protein structure on chromatids where the spindle fibers attach during cell division to pull sister chromatids apart. Their proteins help to hold the sister chromatids together and also play a role in chromosome editing. **1 + 1 = 2**

9. Rudolf Virchow gave this statement. It means "All cells come from cells. Every cell is born of a previous cell, which was born of a previous cell. Life comes from life."

OR

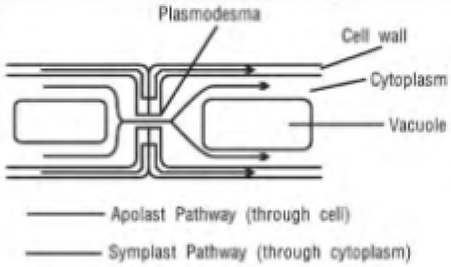
Role of haemoglobin in the transport of respiratory gases :

- Oxygen binds with haemoglobin to form oxyhaemoglobin. 1 molecule of haemoglobin carries 4 molecules of oxygen.
- CO_2 binds with haemoglobin to form carbamino-haemoglobin.CO_2 is carried as carbamino-haemoglobin by blood. **1 + 1**

10.

S.No.	Arithmetic growth	Geometric growth
1.	In arithmetic growth only one daughter cells divides and all the other cells undergo differentiation and maturation.	In geometric growth the growth is proportional to the nutrients supply after which it declines. All the daughter cells divide by mitosis. This is also known as exponential growth.
2.	The graph obtained is a linear one.	Represented by a sigmoid curve where the three phases are initial, exponential and stationary phases.
	Eq : Elongation of root occurring at a constant rate.	**Eq :** Cells growing in a culture medium.

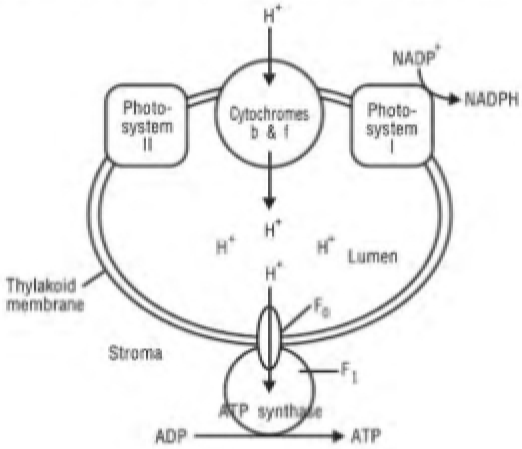
11.



12. Flux refers to the movement of substances across the various compartments. Its two types are:

1. **Flux of movement of molecules through membrane :** It is defined as the rate of transport or diffusion of substances across the semipermeable membrane.
2. **Metabolic flux :** It involves the flow of metabolites through a metabolic pathway. **1 + 2 = 3**

13. **3**



KENDRIYA VIDYALAYA SANGATHAN
[SILCHAR REGION]
SESSION ENDING EXAMINATION 2017
SUBJECT : BIOLOGY
CLASS–XI
(SOLVED PAPER)

Time : 3 Hours **Max. Marks : 90**

General Instructions :

(i) *All questions are compulsory and explain answers with proper headings.*

(ii) *Do not write anything on the questions paper.*

(iii) *Questions No.* ***1*** *to* ***5*** *are of* ***1*** *mark each Answer to these questions carry* ***1*** *mark each. They are required to be answered in one sentence.*

(iv) *Questions No.* ***6*** *to* ***10*** *are of* ***2*** *marks each. Answer to these questions may be answered in about* ***50-25*** *words.*

(v) *Questions No.* ***11*** *to* ***22*** *are of* ***4*** *mark each. Answer to these questions may be answered in about* ***100-120*** *words.*

(vi) *Question No.* ***23*** *is of* ***4*** *marks. Answer to this question may be answered in about* ***150*** *words.*

(vii) *Questions No.* ***24*** *to* ***26*** *are of* ***5*** *marks each. Answer to these question may be answered in about* ***200*** *words.*

(xiii) ***15*** *minutes time has been allotted to read this questions paper.*

1. What is Tidal Volume ? **1**

2. In which type of muscle tissue can you see intercalated discs? What is its significance? **1**

3. Name the pathways by which water flow from roots hair to the deeper layers of root. **1**

4. Why is meiosis known as 'Reduction Division'? What is its significance? **1**

5. Expand the following terms:

(a) ANF (b) JGA **1**

6. List out the major groups of protozoans. **2**

7. Identify the type of synovial joint present in the following parts of the human body: **2**

(a) Between axis and atlas vertebrae

(b) Between humerus and pectoral girdle

(c) Between carpal and metacarpal of thumb

(d) Knee joint

8. What is guttation? Mention the factors responsible for this process. **2**

9. Mention the ploidy of the following:

(a) Protonemal cell of moss

(b) Prothallus cells of fern

(c) Primary androecium nucleus in dicot

(d) Leaf cell of a gymnosperm **2**

OR

What are the rules of binomial nomenclature? **2**

10. Draw the diagram of the following types of chromosomes.

(a) Metacentric (b) Acrocentric **2**

11. What are the various types of nitrogen bases found in DNA? Name the type of bond seen between (a) two nitrogen bases of DNA (b) phosphate and hydroxyl group of sugar of DNA. **3**

12. The ammonia that is absorbed by the plants is used to synthesize various amino acids. Explain the main two ways by which this is done. **3**

13. Identify the function of each of the enzyme given below. Also write the sources of each of these enzymes.

(a) Trypsin (b) Maltase (c) Pepsin **3**

14. Differentiate between C_3 plants and C_4 plants. **3**

15. Draw a well labelled diagram of the regions of the root tip. **3**

16. Identify the phylum that exhibit each of the following feature :

(a) Jointed appendages (b) Comb plates

(c) Water vascular system (d) Pseudocoelom

(e) Metagenesis (f) Notochord **3**

17. Enzymes are classified into different groups based on the reaction they catalyse. Write the name and function of each of these groups of enzymes. **3**

18. Why is ABA known as 'stress hormone'? Mention any two functions of this hormone. How are they antagonistic to gibberellins? **3**

19. How are neurons classified based on the number of axons and dendrites? **3**

OR

Which are the different parts of fore brain? Give any one function of each of these parts. **3**

20. Draw a well labeled diagram of the structure of mitochondrion. **3**

21. The transverse section of a plant material shows the following features:

- Vascular bundles are conjoint, closed and scattered and are surrounded by the sclerenchymatous bundle sheath.

What will you identify it as? Also, write any other four features of this specimen. **3**

22. Draw a labeled diagram of anaphase of mitosis. Write any two key features of this phase. **3**

23. A hospital is arranging an awareness campaign in your locality on 'obesity and other lifestyle disorders'. They are giving a detailed explanation on the causes and effects of these disorders.

(a) As per your opinion, what are the ill effects of Obesity?

(b) What are the various values shown by the hospital authorities in this case?

(c) According to you, who will be benefited by this programme, hospital authority or you. Support your answer by giving reasons.

(d) Name the type of connective tissue that stores fat. **4**

24. Schematically represent the steps of glycolysis. **5**

OR

Explain the steps in non-cyclic photophosphorylation. Why is it called so? **5**

25. Write any five reasons by which a person can become type 2 diabetes patient. **5**

26. How can you distinguish hypothyroidism and hyperthyroidism? Suggest any five ways to distinguish between them. **5**

SOLUTIONS

1. It is the volume of air inspired or expired in each normal breathes. This is about 500 ml in an adult person. It is composed of about 350 ml of alveolar volume and about 150 ml of dead space volume. **1**

2. Cardiac muscles tissue has intercalated discs. Intercalated discs support synchronized contraction of cardiac tissue. **1**

3. Apoplast Pathway and Symplast Pathway. **1**

4. Meiosis is called as reduction division because the chromosome number gets reduced to its half. It helps in producing haploid gametes in sexually reproducing organisms. **1**

5. (a) **ANF –** Atrial Natriuretic Factor

(b) **JGA –** Juxta Glomerular Apparatus. **½ + ½ = 1**

6. Major groups of protozoans are :

(a) Amoeboid Protozoa (b) Flagellated Protozoans

(c) Ciliated Protozoans (d) Sporozoans **½ × 4 = 2**

7. (a) Pivot joint (b) Ball and socket joint

(c) Gliding joint (d) Hinge joint **½ × 4 = 2**

8. Guttation is the process by which excess water is removed in the form of droplets through hydathodes. Root pressure and transpiration are the factors responsible for this process. **1 + 1 = 2**

9. (a) Protonemal cell of a moss – Haploid

(b) Prothallus of a fern – Haploid

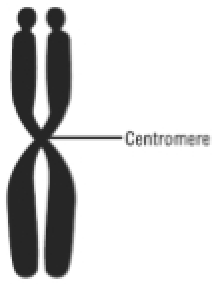
(c) Primary androecium nucleus in dicot – diploid

(d) Leaf cell of a gymnosperm – Haploid **½ × 4 = 2**

OR

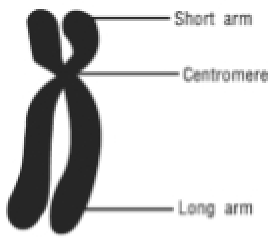
Every scientific name has a first word for genus followed by a second word denoting the species. Hence, binomial nomenclature is the system of scientific naming of an organism using 'genus' as the first part and 'species' as the second part as stated by Carolus Linnaeus. **2**

10. (a)



Metacentric chromosome

(b)



Acrocentric chromosome

1 + 1 = 2

11. Types of nitrogenous bases found in DNA are:

(i) **Adenine :** abbreviated 'A,' has a 2-ring structure, so that makes it a purine.

(ii) **Thymine :** abbreviated 'T,' is a pyrimidine, which means it has a 1-ring structure. It's only present in DNA, where it pairs with adenine.

(iii) **Guanine :** abbreviated 'G,' is part of both DNA, where it bonds with cytosine.

(iv) **Cytosine :** abbreviated 'C,' is part of DNA and bonds with guanine. It has one ring, so it's a pyrimidine.

(a) Hydrogen bond

(b) Phosphodiester bond **3**

12. At physiological pH, the ammonia is protonated to form NH_4^+ (ammonium) ion while most of the plants can assimilate nitrate as well as ammonium ions, the latter is quite toxic to plants and hence cannot accumulate in them.

KENDRIYA VIDYALAYA SANGATHAN
[AGRA REGION]
SESSION ENDING EXAMINATION 2017
SUBJECT : BIOLOGY
CLASS–XI
(SOLVED PAPER)

Time : 3 Hours **Max. Marks : 90**

General Instructions :

*(i) Answers to questions carrying **1** mark may be answered in one word to one sentence.*
*(ii) Answers to questions carrying **3** marks may be answered in about **50-75** words.*
*(iii) Answer to questions carrying **4** marks may be answered in about **150** words.*
*(iv) Answers to questions carrying **6** marks may be answered in about **200** words.*
(v) Attempt all parts of a question together.

1. What is the nature of the cell wall? **1**

2. Name the substance that fixes the brown algae to the substrate. **1**

3. Name the site of synthesis of glycoproteins and glycolipids in a cell. **1**

4. How many types of amino acids found to occur in proteins? **1**

5. What is the value of water potential of pure water? **1**

6. Bring out the differences between chlorophyceae and phaeophyceae. **2**

7. Give the location of hepatic caecae in a cockroach. What is their function? **2**

8. What are kinetochores? What is their function? **2**

9. *'Omnis cellula-e-cellula'*. Who gave this statement and what does it mean? **2**

OR

Describe the role of haemoglobin in the transport of respiratory gases. **2**

10. Write differences between arithmetic and geometric growth rate. **2**

11. Represent diagrammatically the apoplastic and symplastic pathways of water transport in plants. **3**

12. What is meant by flux? Describe its two kinds.

13. Represent systematically the process of ATP synthesis through chemiosmosis in chloroplast. **3**

14. Give the structural formula of (a) adenine (b) adenosine (c) adenylic acid

15. Draw a well labeled diagram of mitochondria. **3**

16. Where does mitosis take place in plants and animals? What is its significance in multicellular organisms? **3**

17. Represent diagrammatically a pinnately compound leaf and a palmately compound leaf. Name one example of each. **3**

18. What is periderm? How does periderm formation takes place in dicot stem? **3**

19. Draw a well labeled diagram of alimentary canal of a cockroach. **3**

20. Who proposed the five kingdom classification? Name the five kingdoms. **3**

OR

How are the animals of Arthropoda different from those of Mollusca? **3**

21. What is exophthalmic goitre? Mention any four symptoms of this disorder. **3**

22. Write short notes on hind brain. **3**

23. Leena and Rita are final year BSc. students. Suddenly their classmate Arun's mother needed blood transfusion. She is B positive. Leena and Rita made all efforts to find suitable blood donor from their college to help Arun.

(a) What blood group/groups can be transfused by Arun's mother.

(b) Name the blood group known as universal donor.

(c) Name the blood group referred to as universal acceptor.

(d) What values are shown by Leena and Rita? **4**

24. Where does non cyclic photophosphorylation take place? Describe this process with the help of its schematic representation. Why is this process called so? **5**

OR

Where does glycolysis occur in cell? Describe the sequence of reactions in it. Mention the end product. **5**

25. Suggest five measures that can be proposed at global level for the prevention and control of diabetes and its complications. **5**

26. Mention the schemes implemented by government of India to tackle malnutrition and health problems in children. **5**

SOLUTIONS

1. The plant cell wall is a complex matrix of linked polysaccharides such as cellulose and pectin, forming a thick semi-permeable rigid barrier outside the plasma membrane.

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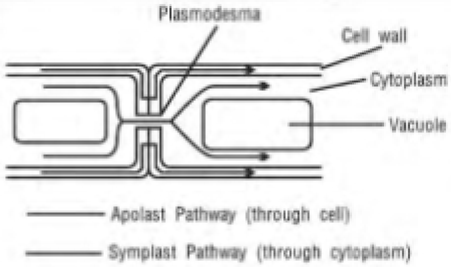

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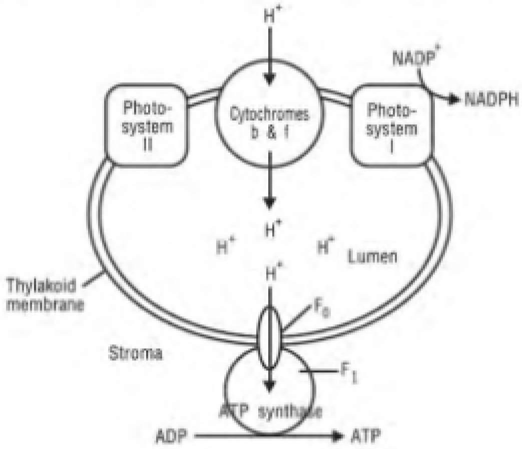


ST. XAVIER'S SENIOR SECONDARY SCHOOL, DELHI - 110 054
Annual Examination in **BIOLOGY**

Std. 11 | Time : 3 hrs.
3-3-2018 | Max. Marks : 70

General Instructions :-

i) All questions are compulsory.
ii) This question paper consists of five Sections A, B, C, D and E. Section A contains 5 questions of one mark each, Section B is of 5 questions of two marks each, and Section C is of 12 questions of three marks each. Section D is of 1 question of four marks and Section E is of 3 questions of five marks each.
iii) There is no overall choice. However, an internal choice has been provided in one question of 2 marks, one question of 3 marks and all the three questions of 5 marks weightage. A student has three questions of 5 marks weightage. A student has to attempt only one of the alternatives in such questions.
iv) Wherever necessary, the diagrams drawn should be neat and properly labelled.

SECTION – A

1. Why is binomial nomenclature the most acceptable mode of naming organism? [1]

2. What is functional residual capacity? [1]

3. Name the secretions of Goblet cell & parietal cells. [1]

4. How open vascular bundles are differing from closed vascular bundles? [1]

5. Who first explained that new cells arise from pre-existing cells? [1]

SECTION – B

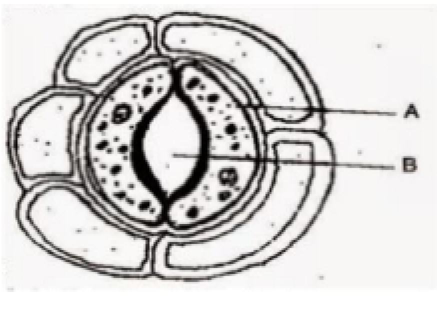
6. What is the difference between carbaminohaemoglobin and oxyhaemoglobin? [2]

7. How is transpiration different from guttation? Give two points. [2]

8. Fill in the blank spaces in the given table to bring the difference between C3 and C4 plants: [2]

	C3 plants		C4 plants
i)	Cell type: One type (mesophyll)	a)	________________
ii)	CO_2 acceptor ________________	b)	Phosphoenol pyruvate (PEP)
iii)	First CO2 fixation Product: 3-PGA	c)	________________
iv)	Optimum temperature __________	d)	30° C to 45° C

9. Observe the figure and answer the following questions : [2]

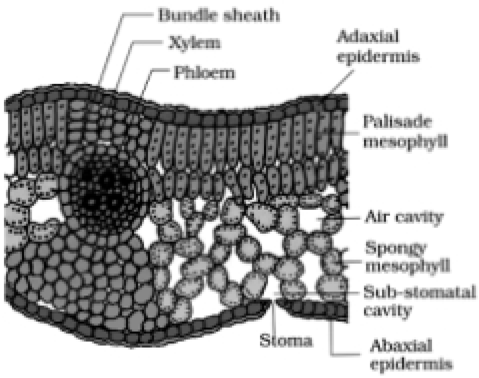


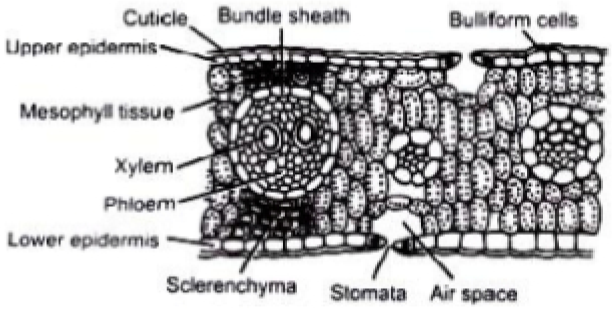
a) Name parts (a) and (b).
b) Are these types of stomata observed in monocot or in dicot plants?
c) Which parts of stomata constitute the stomatal apparatus?

10. Name the type of root for the following : [2]
   a) Roots performing the function of photosynthesis.
   b) Roots come above the surface of the soil to absorb air.
   c) The pillar like roots developed from lateral branches for providing mechanical support.
   d) Roots coming out of the lower nodes of the stem and provide the support to the plant.

(OR)

Identify and differentiate between the following :-





SECTION – C

11. Where are synaptic vesicles found? Name their chemical contents? What is the function of these contents? [3]

12. a) A patient was complaining of frequent urination, excessive thirst, hunger and tiredness. His fasting glucose level was found higher than 130 mg/dL on two occasions :
      i) Name the disease,
      ii) Give the root cause of this disease.
      iii) Explain why the blood glucose level is higher than 130 mg/dL.
   b) Where will you find the following?
      i) Nephron ii) Fovea iii) Organ of Corti.
      Give one function of each. [3]

13. Differentiate between arteries and veins. [3]

14. Explain the different types of phyllotaxy. Give one example of each type. [3]

15. Give the schematic representation of an overall view of TCA cycle.

(OR)

Schematically represent non-cyclic photophosphoryaltion in plants. How is it different from the cyclic photophosdhorylation? [3]

16. Give steps of ATP synthesis in chloroplasts through chemiosmosis. [3]

17. Explain the Fluid Mosaic Model. Also represent it diagrammatically. [3]

18. a) Differentiate between the cytokinesis of plant and animal cell. [2]
   b) What is G0 phase of cell cycle? [1]

19. Differentiate between a prokaryotic and a eukaryotic cell. [3]

20. State the Blackman's law of the limiting factors. Explain the effect of light and carbondioxide on photosynthesis. [3]

21. Differentiate between the following:- [3]
   a) Polyp and medusa.
   b) Pseudocoelomates and acoelomates.
   c) Oviparous and viviparous.

22. Give Reasons for the following :- [3]
   i) Ferns reproduce by their leaves.
   ii) Fungal hypha shows the presence of two nuclei per cell in some Basidiomycetes.
   iii) Mycorrhizal association is advantageous to the plants.

SECTION – D

23. Class XI went for an excursion trip and they saw different structures described as stems by their teacher. How do plants support, store, multiply and protect the plants by their stems? Give examples of each category to explain them. [4]

SECTION – E

24. Draw a detailed diagram of a plant cell and differ it with the animal cell.

(OR)

Explain the following:- [5]
   a) The phenomenon of crossing over.
   b) Anaphase of Mitosis and Meiosis.
   c) Plastids.

25. Describe the structure of human heart. Draw the diagram and show the path of blood circulation through it.

(OR)

Draw a labelled diagram of human alimentary canal and describe the role of the following:-
   a) Bile b) Pepsin [5]

26. How do plants absorb water? Explain transpiration pull model in this regard.

(OR)

   a) Describe the pressure flow hypothesis of translocation of sugar in plants.
   b) Explain the mechanism of closing and opening of stomata. [5]

-x-x-x-x-x-x-x-

ROLL NO.	
NAME	
CLASS & SECTION	

APEEJAY COMMON ANNUAL EXAMINATION, 2018-19 **02**

CLASS-XI

BIOLOGY (044)

Time allowed : 3 hrs. *Maximum Marks : 70*

General Instructions :

(i) *There are a total of 27 questions and four sections in the question paper. All questions are compulsory.*

(ii) *Section A contains question numbers 1 to 5, Very Short answer type questions of one mark each.*

(iii) *Section B contains question numbers 6 to 12, Short Answer type I questions of two marks each.*

(iv) *Section C contains question numbers 13 to 24, Short Answer type II questions of three marks each.*

(v) *Section D contains question numbers 25 to 27, Long Answer type questions of five marks each.*

(vi) *There is no overall choice in the question paper; however, an internal choice is provided in two questions of one and two marks each, three questions of three marks and all questions of five marks. An examinee is to attempt any one of the questions out of the two given in the question paper with the same question number.*

SECTION-A

1. What will be the axis of expansion of the thoracic chamber when external intercostal muscles and diaphragm contracts during inspiration? (1)

2. Which organ is affected when a person is suffering from jaundice and why does skin and eyes turn yellow? (1)

3. If you are given twigs of two plants how will identify a twig having simple leaf from that of a twig having compound leaves? (1)

4. What is the advantage of having more than one pigment molecules in a Photocenter? (1)

OR

How chloroplasts are aligned in the mesophyll cells during light reaction and why do they do so?

5. Why an oxygen scavenger, leg-haemoglobin is required in the root nodules of leguminous plants?

OR

Why flow of water in the roots via apoplastic way beyond endodermis impossible? (1)

SECTION-B

6. Two potted plants were kept in an oxygen free environment in transparent containers, one in total darkness and the other in sunlight. Which one of the two is likely to survive more? Justify your answer by giving the reason.

OR

Explain the type of photophosphorylation that occurs when light of wavelengths beyond 680 nm are available for excitation. (2)

7. How is chemical analysis of organic compounds in a living tissue carried out? (2)
8. Draw a diagram of a lenticel and label any four of its components. (2)
9. What are Cnidoblasts? Where are they present? What are they useful for? (2)

OR

Using a high precision microscope, how will you distinguish between a chrysophyte from a dinoflagellate ?

10. What effect will be observed in the foetus of a pregnant lady suffering from hypothyroidism? (2)
11. Explain how crossing over is initiated and completed at the end of pachytene stage. (2)
12. How are herbarium sheets prepared? What information does the label in the herbarium sheet provide for taxonomical studies? (2)

SECTION-C

13. (a) What is compound epithelium? What are their main functions?
    (b) Where do we find areolar tissue?
    (c) How is adhering junction different from gap junction? (3)

14. If the mesophyll part of cut vertical sections of leaves belonging to C3 and C4 plant are studied what differences will be observed? State any three of them. (3)

15. (a) How many cells are required to produce 200 cells at the end of meiotic division?

(b) The number of bivalents in a cell at Prophase I stage of meiosis is 10, how many chromosomes will be present in Metaphase I, Anaphase I and anaphase II? Give reason. (3)

16. With the help of a diagram explain the structure of actin.

OR

Explain how joints are classified into three major structural forms. (3)

17. Explain with the help of an example, as to how the presence of certain chemicals that bind to the enzyme, shuts off enzyme activity and what are such chemicals called? (3)

18. Name the mineral that is responsible for the following :

(a) Maintenance of ribosomal structure

(b) Splitting of water to liberate oxygen during photolysis

(c) Synthesis of Auxin

(d) Germination of pollen grain

(e) Nitrogen metabolism

(f) Anion, that helps in maintenance of anion-cation balance

OR

Describe the following three deficiency symptoms and co-relate them with concerned mineral deficiency :

(a) Chlorosis

(b) Necrosis

(c) Stunted plant growth (3)

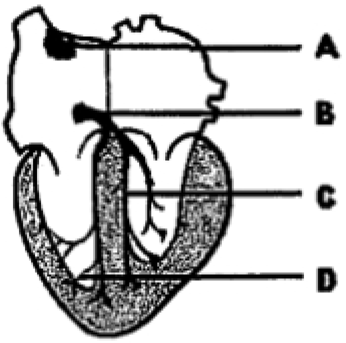
19. Define the term and explain the conditions that lead to water loss in liquid phase in plants? (3)

20. How the functioning of the kidney is efficiently monitored and regulated by JGA?

OR

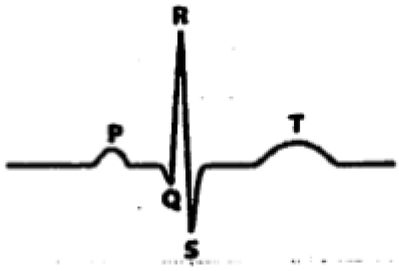
Explain the mechanism by which human kidneys are capable to produce urine nearly four times concentrated than the initial filtrate. (3)

21. The diagram that is depicted below is the conducting system of the heart, name the parts labelled as A, B, C and D and also mention the role of A and B.



OR

Explain how standard ECG shown below helps in the evaluation of the heart function. (3)



22. Mention the role played by :

    (a) Long thin petiole of a leaf

    (b) Veins

    (c), Petiole of Australian acacia (3)

23. The stomatal pore is guarded by two guard cells. Name the epidermal cells surrounding the guard cells. How does a guard cell differ from an epidermal cell on the basis of shape and thickness of wall? (3)

24. (a) How Mosses along with lichens are of great ecological importance.

    (b) Apart from being commercially viable, how are Algae useful to food chain and environment? (3)

SECTION-D

25. (a) It was found that some varieties of wheat which were sown in spring failed to flower or produce mature grain within a span of a flowering season. What could be the probable reason for this, elaborate and also name the phenomeno. (5)

    (b) Elaborate the role of phytohormones in the following :

    (i) to prepare weed-free lawns by gardeners

    (ii) to increase the length of grapes stalks

    (iii) to enable hedge-making.

OR

(a) Mention the two crucial events taking place in aerobic respiration. Why the role of oxygen is limited to the terminal stage of the aerobic process of respiration ?

(b) Explain how energy released during the electron transport system is utilized in synthesizing ATP with the help of ATP synthetase.

26. Name the parts of the alimentary canal where absorption of digested products take place in humans and give a detailed account on the various digested products that are absorbed in those parts? (5)

OR

(a) What are the areas of control exercised by hypothalamus and what are the other regulatory mechanisms that the hypothalamus controls in association of limbic system?

(b) How transmission of nerve impulses is different in an electrical synapse and in a chemical synapse?

27. (a) Explain the structure and role of the nucleolus and how chromatin material are structurally modified during different stages of cell division. Name the organelles that do not show during cell division when viewed under the microscope.

(b) What is the significance of the nuclear envelope being interrupted at various places? (5)

OR

(a) How are proteins classified on the basis of ease of extraction on the membrane?

(b) How are neutral solutes and polar molecules transported across the membranes?

(c) Mention the features that are common in Chloroplasts, Mitochondria and prokaryotic cells.

BEST OF LUCK!

ROLL NO.	
NAME	
CLASS & SECTION	

APEEJAY COMMON ANNUAL EXAMINATION, 2018-19 02

CLASS-XI

BIOLOGY (044)

Time allowed : 3 hrs. *Maximum Marks : 70*

General Instructions :

(i) There are a total of 27 questions and four sections in the question paper. All questions are compulsory.

(ii) Section A contains question numbers 1 to 5, Very Short answer type questions of one mark each.

(iii) Section B contains question numbers 6 to 12, Short Answer type I questions of two marks each.

(iv) Section C contains question numbers 13 to 24, Short Answer type II questions of three marks each.

(v) Section D contains question numbers 25 to 27, Long Answer type questions of five marks each.

(vi) There is no overall choice in the question paper; however, an internal choice is provided in two questions of one and two marks each, three questions of three marks and all questions of five marks. An examinee is to attempt any one of the questions out of the two given in the question paper with the same question number.

SECTION-A

1. What will be the axis of expansion of the thoracic chamber when external intercostal muscles and diaphragm contracts during inspiration? (1)

2. Which organ is affected when a person is suffering from jaundice and why does skin and eyes turn yellow? (1)

3. If you are given twigs of two plants how will identify a twig having simple leaf from that of a twig having compound leaves? (1)

4. What is the advantage of having more than one pigment molecules in a Photocenter? (1)

OR

How chloroplasts are aligned in the mesophyll cells during light reaction and why do they do so?

5. Why an oxygen scavenger, leg-haemoglobin is required in the root nodules of leguminous plants?

OR

Why flow of water in the roots via apoplastic way beyond endodermis impossible? (1)

SECTION-B

6. Two potted plants were kept in an oxygen free environment in transparent containers, one in total darkness and the other in sunlight. Which one of the two is likely to survive more? Justify your answer by giving the reason.

OR

Explain the type of photophosphorylation that occurs when light of wavelengths beyond 680 nm are available for excitation. (2)

7. How is chemical analysis of organic compounds in a living tissue carried out? (2)

8. Draw a diagram of a lenticel and label any four of its components. (2)

9. What are Cnidoblasts? Where are they present? What are they useful for? (2)

OR

Using a high precision microscope, how will you distinguish between a chrysophyte from a dinoflagellate ?

10. What effect will be observed in the foetus of a pregnant lady suffering from hypothyroidism? (2)

11. Explain how crossing over is initiated and completed at the end of pachytene stage. (2)

12. How are herbarium sheets prepared? What information does the label in the herbarium sheet provide for taxonomical studies? (2)

SECTION-C

13. (a) What is compound epithelium? What are their main functions?

    (b) Where do we find areolar tissue?

    (c) How is adhering junction different from gap junction? (3)

14. If the mesophyll part of cut vertical sections of leaves belonging to C3 and C4 plant are studied what differences will be observed? State any three of them. (3)

15. (a) How many cells are required to produce 200 cells at the end of meiotic division?

(b) The number of bivalents in a cell at Prophase I stage of meiosis is 10, how many chromosomes will be present in Metaphase I, Anaphase I and anaphase II? Give reason. (3)

16. With the help of a diagram explain the structure of actin.

OR

Explain how joints are classified into three major structural forms. (3)

17. Explain with the help of an example, as to how the presence of certain chemicals that bind to the enzyme, shuts off enzyme activity and what are such chemicals called? (3)

18. Name the mineral that is responsible for the following :

(a) Maintenance of ribosomal structure

(b) Splitting of water to liberate oxygen during photolysis

(c) Synthesis of Auxin

(d) Germination of pollen grain

(e) Nitrogen metabolism

(f) Anion, that helps in maintenance of anion-cation balance

OR

Describe the following three deficiency symptoms and co-relate them with concerned mineral deficiency :

(a) Chlorosis

(b) Necrosis

(c) Stunted plant growth (3)

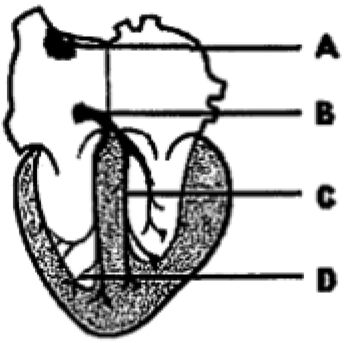
19. Define the term and explain the conditions that lead to water loss in liquid phase in plants? (3)

20. How the functioning of the kidney is efficiently monitored and regulated by JGA?

OR

Explain the mechanism by which human kidneys are capable to produce urine nearly four times concentrated than the initial filtrate. (3)

21. The diagram that is depicted below is the conducting system of the heart, name the parts labelled as A, B, C and D and also mention the role of A and B.



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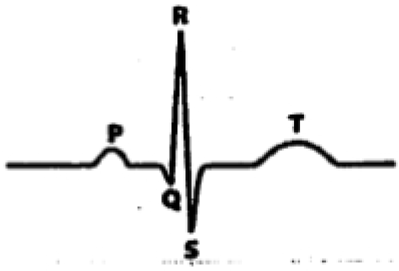

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SECTION-D

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BEST OF LUCK!

Roll No.	
Name	
Class & Section	

APEEJAY COMMON ANNUAL EXAMINATION, 2019-20

BIOLOGY (044)

Time Allowed : 3.00 Hrs. **Class – XI** **Maximum Marks : 70**

General Instructions :

(1) There are a total of 27 questions and five sections in the question paper. All questions are compulsory.

(2) Section A contains question numbers 1 to 5, multiple choice questions of one mark each. Section B contains question numbers 6 to 12, short answer type I questions of two marks each. Section C contains question numbers 13 to 21, short answer type II questions of three marks each. Section D contains question number 22 to 24, case-based short answer type questions of three marks each (1+1+1).

Section E contains question numbers 25 to 27, long answer type questions of five marks each.

(3) There is no overall choice in the question paper. However, internal choices are provided in two questions of one mark, one question of two marks, two questions of three marks and all three questions of five marks. An examinee is to attempt any one of the questions out of the two given in the question paper with the same question number.

SECTION-A

1. The embryo sac of an angiosperm is made up of : **(1)**

   (a) 7 cells and 7 nuclei (b) 8 cells and 7 nuclei

   (c) 7 cells and 8 nuclei (d) 8 cells and 8 nuclei

2. Two common characters found in centipede, cockroach and crab are : **(1)**

   (a) Jointed legs and chitinous exoskeleton

   (b) Green glands and tracheae

   (c) Book lungs and antennae

   (d) Compound eyes and anal cerci

OR

A list of animals is given below. Identify the animals with open circulatory system and choose the correct answer:

(i) Nereis
(ii) Cockroach
(iii) Earthworm
(iv) Prawn

(a) (i) and ii)
(b) (ii) and iv)
(c) (ii) and m)
(d) (iii) and iv)

3. Leg-haemoglobin helps in : **(1)**

(a) Nitrogen fixation
(b) Protecting Nitrogenase from O_2
(c) Destroys bacteria
(d) Transport of food in plants

OR

The process in which water is lost in the form of liquid droplets :

(a) Guttation
(b) Transpiration
(c) Evaporation
(d) Osmosis

4. Which of the following is true regarding glycolysis : **(1)**

(i) Takes place in cytosol

(ii) Produces no ATP

(iii) Has no connection with electron transport chain

(iv) Reduces two molecules of NAD+ for every glucose

Choose the correct option :

(a) Only (i)
(b) (i), (ii) and (iii)
(c) (i) and (ii)
(d) none of these

5. Organic compounds which contain an amino group and an acidic group as substituents on the alpha carbon. **(1)**

(a) Carbohydrates
(b) Proteins
(c) Amino acids
(d) Nulceic Acids

SECTION-B

6. Heterospory is the characteristic feature of the life of few Pteridophytes. (2)

   (a) State the evolutionary significance of this in the plant kingdom.

   (b) Which development led to this very significant step in evolution?

7. Why Mitosis is called equational cell division? Where does it occur?

8. After keeping a freshly collected Spirogyra filament in 10% sodium chloride solution, it is observed that protoplasm shrinks in size. (2)

   (a) Name the phenomenon occurring in the above case?

   (b) What will happen if filament is kept in distilled water?

9. Define dedifferentiation and redifferentiation. (2)

10. Explain the heart sounds produced during a cardiac cycle. (2)

11. Briefly explain the process of digestion of fats in the small intestine. (2)

12. The pituitary gland is divided anatomically into two parts. Name one hormone secreted by each of these parts and mention the function of each. (2)

OR

Name the hormones secreted by α-cells and β-cells of the pancreases. How is glucose homeostasis in our blood maintained by these hormones?

SECTION-C

13. Give a comparative account of the classes of Kingdom Fungi under the following :

    (a) Mode of nutrition

    (b) Mode of reproduction

    (c) Mycelium

OR

Name the three divisions of Algae and compare them on the basis of :

(a) Major pigments (b) Stored food

(c) Cell wall

14. Represent diagrammatically the (a) coelomate (b) acoelomate and (c) pseudocoelomate conditions among animals ? (3)

OR

(a) What are the differences between a virus and a viroid?

(b) Define 'symbiots' and give an example of the same.

15. In nearly all animal tissues, specialized junctions provide structural and functional links. How many types of such junctions are present? List their names and state their functions. (3)

16. State how placenta is related to female reproductive part in flowering plants. Mention the types of placentation found in the angiosperms. (3)

17. How do enzymes speed up the rate of a chemical reaction? Explain using a suitable graphical representation. (3)

18. Explain 9+2 arrangement of Cilium. Support your answer with a labelled diagram. (3)

19. What is respiratory quotient? Compare its value for carbohydrates, proteins and lipids? Which out of the three respiratory substrates have maximum RQ and why? (3)

20. Draw a neat sectional view of the heart and label the following parts : (3)

(a) Nodal tissue from where the heart-beat generates.

(b) Chordae tendinae

(c) An artery which carries deoxygenated blood.

(d) Bundle of His

21. Write two points of differences between : (3)

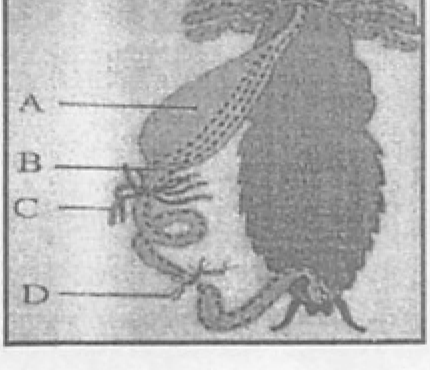
(a) Actin and myosin

(b) Red and white muscles

(c) Pectoral and Pelvic girdle

SECTION-D

22.



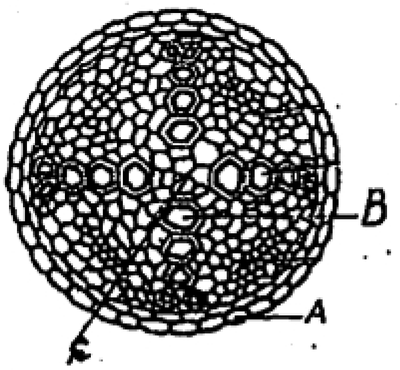
Observe the given diagram :

(a) Identify the parts A and D

(b) Discuss the role of part B in the process of digestion.

(c) What function does the part D has?

23.



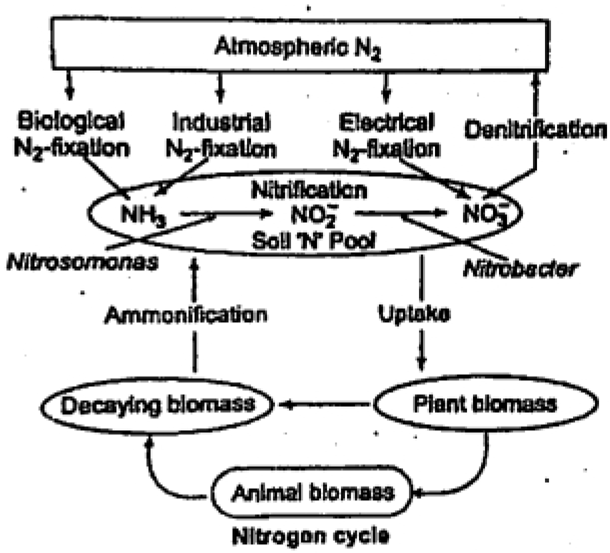
A class XI student made a temporary preparation of T.S. of sunflower root and observed it under the microscope and drew the given diagram.

(a) What is part A depicting ? List its one characteristic feature.

(b) Name parts B and C

(c) What type of arrangement does xylem show? Name this type of arrangement.

24.



Nitrogen cycle

24. Nitrogen is an important nutrient for all living organisms. It forms amino acids and proteins. It occurs as N_2 in atmosphere and cannot be directly utilised. Plants compete with microbes for the limited nitrogen that is available in the soil. And hence nitrogen becomes a limiting nutrient for both natural and agricultural ecosystems.

(a) State the methods by which atmospheric nitrogen can be fixed and used by plants.

(b) Name one bacterium which reduces nitrate back into N_2 .What is this process called?

(c) Name the enzyme which is exclusively present in prokaryotes and is essential for nitrogen fixation.

SECTION-E

25. (i) How do neutral solutes move across the plasma membrane? Can the polar molecules also move across it in the same way? If not, how are these transported? What is the requirement for carrying ions across the membrane?

(ii) Give specific term for each of the following :

(a) Cluster of ribosomes found in cytoplasm.

(b) Intensive infolding of the inner membrane of mitochondria.

(c) Stacks of closely packed thylakoids.

(d) Stalked particles on the inner membrane of mitochondria.

OR

(i) Briefly explain the Metaphase, Anaphase and Telophase of mitosis with the help of diagrams.

(ii) Explain the following terms :

(a) Synaptonemal complex (b) Chiasmata

26. Draw the diagrammatic representation of Hatch and Slack pathway. What type of plants show this pathway? Give one example. Their leaf anatomy) is different from other plants. How?

OR

Draw schematic representation of the various steps of Glycolysis. How is glycolysis different from citric acid cycle? State two points.

27. Draw and explain the schematic representation of generation and conduction of nerve impulse from point A to point B.

OR

With the help of diagrammatic representation, explain the mechanism of hormone action :

(a) Protein hormone (b) Steroid hormone